

ANALYSIS OF SYMBOLIC MUSIC & COMPUTATIONAL MUSICOLOGY

Monteverdi's Madrigals – A Statistical Analysis

Prima Pratica, Seconda Pratica and expressing Sprezzatura

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Can Stylistic Evolution Be Quantified?

1

Stylistic Evolution

Computationally track Claudio Monteverdi's musical language across all 9 books of madrigals using symbolic analysis tools.

2

Quantify the Pratiche

Measure the shift from Renaissance counterpoint (*Prima Pratica*) to text-driven harmony (*Seconda Pratica*) in concrete data.

3

Formalize Sprezzatura

Translate the abstract Renaissance concept of "effortless grace" into measurable, computable musical parameters.

Can Stylistic Evolution Be Quantified?

What is a Madrigal?

A refined secular vocal form for typically 5 independent voices, emerging c. 1530. The "chamber music" of the Renaissance: poetic text directly dictated musical gesture (*madrigalisms*).

Renaissance Context (1500–1600)

Dense polyphony governed by strict counterpoint rules, gradually evolving toward theatrical monody and individual vocal expression.

Can we computationally formalize the shift from *Prima* to *Seconda Pratica* using symbolic music data?

The Goals

- Create a Dataset from Monteverdi's books
 - Analyze symbolic differences between *Prima* and *Seconda Pratica*
 - Formalizing *Sprezzatura* from data

The Primary Conflict: Two Practices

Prima Pratica – The Conservative Baseline

Traditional counterpoint governs harmony as master over text. Rules are inviolable; dissonance must be prepared and resolved.

Artusi's Backlash (1600)

"...introduce nuove Regole, nuovi modi e nuova frase nel dire, sono però aspri, e all'udito poco piacevoli..."

— *L'Artusi overo le imperfettioni della moderna musica*

Seconda Pratica – The Manifesto

Giulio Cesare Monteverdi's, Claudio's brother, response reframes the entire hierarchy of musical elements:

"By First Practice he means that which turns on the perfection of the harmony, that is, which considers harmony not commanded but commanding, and not the servant but the mistress of the oration (...): By Second Practice (...) he means that which turns on the perfection of the melody, that is, which considers harmony commanded and not commanding, and makes the oration the mistress of the harmony."

Cruda Amarilli (Guarini, Il pastor fido, Atto I, scena 2, 1590)

Italiano (Original):

Cruda Amarilli, che col nome ancora,
d'amar, ahì lasso, amaramente insegni;
Amarilli, del candido ligustro
più candida e più bella,
ma de l'aspido sordo
e più sorda e più fera e più fugace,
poi che col dir t'offendo,
i' mi morrò tacendo.

English Translation:

Cruel Amarilli, who with your very name
teaches me, alas, to love bitterly;
Amarilli, than the white privet flower,
more white and more beautiful
but more deaf than the deaf asp,
and more fierce and more fleeting,
since speaking offends you,
I shall die in silence.



Why It Matters

Artusi singled out this very madrigal as evidence of Monteverdi's "imperfections." It is our primary computational test subject.

Cruda Amarilli

Dissonàncies de 7a i 9a preparada per moviment oblic i grau conjunt entre el Basso i el Canto. Resolen baixant una 2a desc. Emfatitzat en produir-se entre veus extremes. Simbolitza el lament del protagonista (Mirtillo) en dir Cruda (cruel) i nombrar la seua estimada Amarilli.

Cruda Amarilli

1a PART: Queixes

Claudio Monteverdi

Canto

Alto

Quinto

Tenore

Basso

Fel - - - le a - ma - - ro, fel - - - le a - ma - -
Cru - - - da A - ma - ril - - li, cru - - - da A - ma - ril - - li

Alumnes de GP5 del Conservatori de Tortosa. Tortosa, 29 d'octubre de 2015.

Guarini's Text

Cruda = cruel. The word itself triggers harmonic violation.

Amarilli contains the roots of both *amare* (to love) and *amaro* (bitter), a play of words that Monteverdi emphasized with the composition.

Dissonance on 'Cruda Amarilli'



Cruda Amarilli

Dissonàncies sense preparació després de silenci. Simbolitza el lament desesperat de l'enamorat.

Fragment homofònic per expressar amb claredat el missatge del text: que ella l'ha ensenyat a estimar.

13

C

A

Q

T

B

- ce - - - to,
las - - - so,

me po - ta - uit po - pu - lus &
che col no-me an-co - ra d'a-mar

a - - - ce - - -
ahi!

- ce - - - to,
las - - - so,

me po - ta - uit po - pu - lus &
che col no-me an-co - ra d'a-mar

a - - - ce - - -
ahi!

- ce - - - to,
las - - - so,

me po - ta - uit po - pu - lus &
che col no-me an-co - ra d'a-mar

a - - - ce - - -
ahi!

Guarini's Text

*"che col nome ancorad'amar, ah! lasso,
amaramente insegna"*

"who with your name still teaches me to love bitterly"

Dissonance on 'Che col nome ancora'



From Concept to Research Gap



Castiglione (1528)

Sprezzatura defined as effortless, artful grace, concealing the labor behind mastery.



Caccini (1602)

"Noble negligence of singing" —
Tempo dell'animo (soul's time) against
Tempo della mano (strict beat).



The Gap

Quantitative analysis using computational tools to determine presence of *Sprezzatura* remains absent.

Dataset Curation & Tooling

Data Source

Choral Public Domain Library (ChoralWiki) — representative madrigals from all 9 books.

Pipeline

Scraped, cleaned, and synchronized each madrigal piece into a triad of formats.

Format Triad

PDF

Visual ground-truth of the score.

MIDI

Symbolic representation, useful for future applications.

MusicXML

Core symbolic data for object-oriented analysis

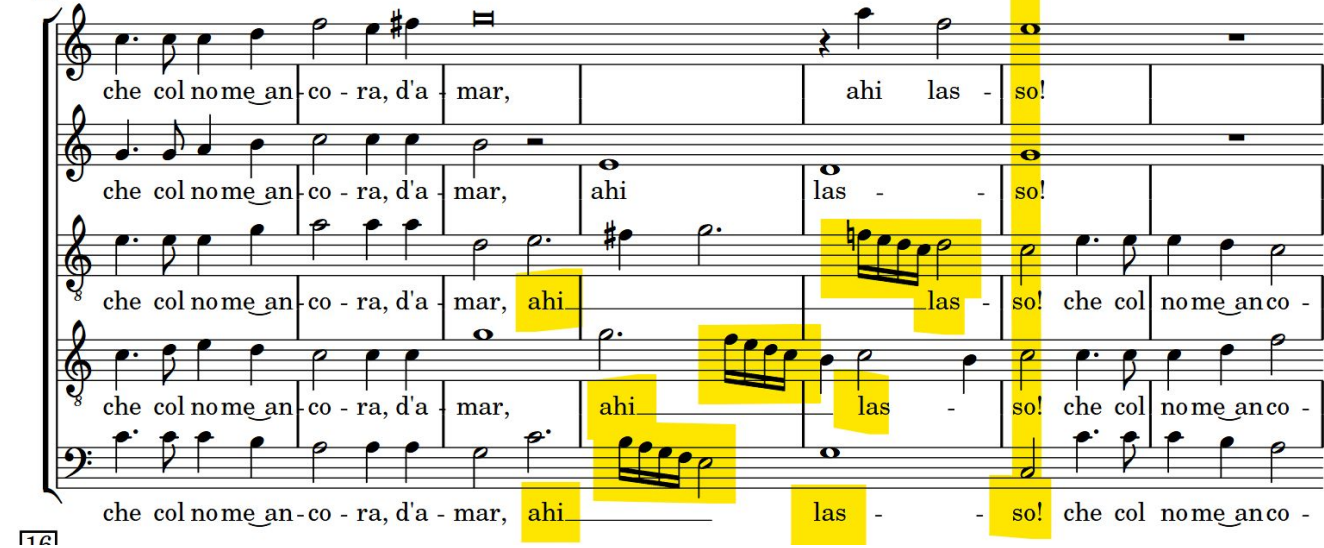
Python

Statistical modeling & scripting

music21

Symbolic music parsing, interval analysis, rhythmic extraction

Quantifying the **Pratiche**



Seconda Pratica has its starting point in the fifth book. In which Monteverdi's philosophical ideas were first applied in both music and the text.

1

Dissonance Ratio

Ratio of **unprepared dissonances** (7ths, 9ths on strong beats) against total harmonic intervals per book.

2

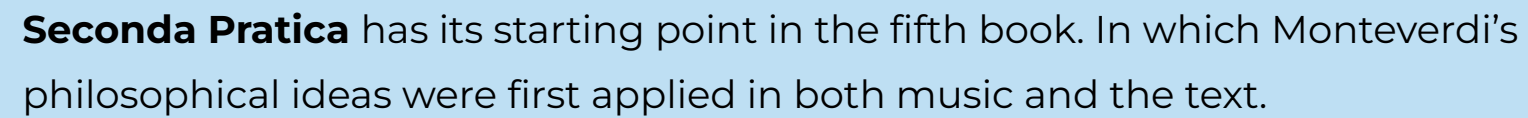
Textural Evolution

Percentage of **homorhythmic behavior** (chordal, together) versus independent polyphonic voice-leading.

3

Basso Continuo

Appearance of a harmonic anchor designed specifically to support and liberate the emotional expression.



!

Monteverdi argued that the words must be the "master of the harmony, not the servant."

Dissonance and Text

How was dissonance quantified per book?

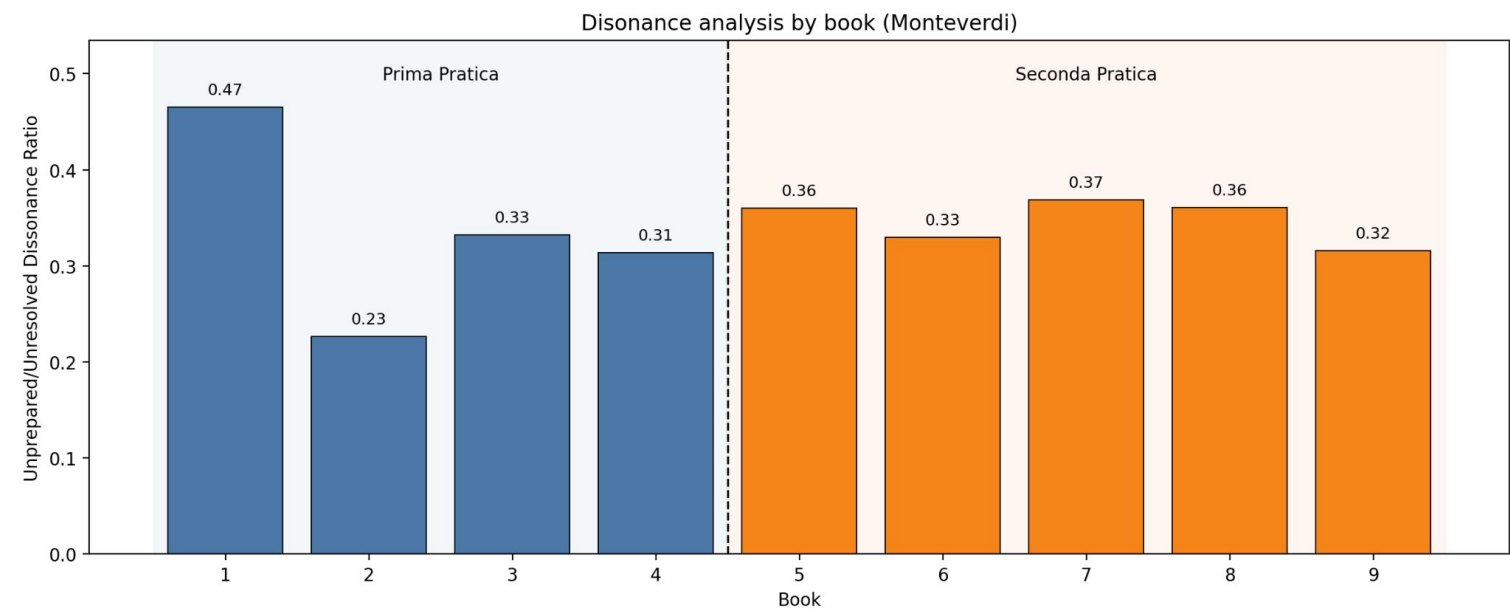
Analyzing dissonant interval against any other simultaneously sounding voice. Those intervals are set as minor and major 2nd, tritone, minor and major 7th.

$$\text{DISSONANT_MOD12} = \{1, 2, 6, 10, 11\}$$

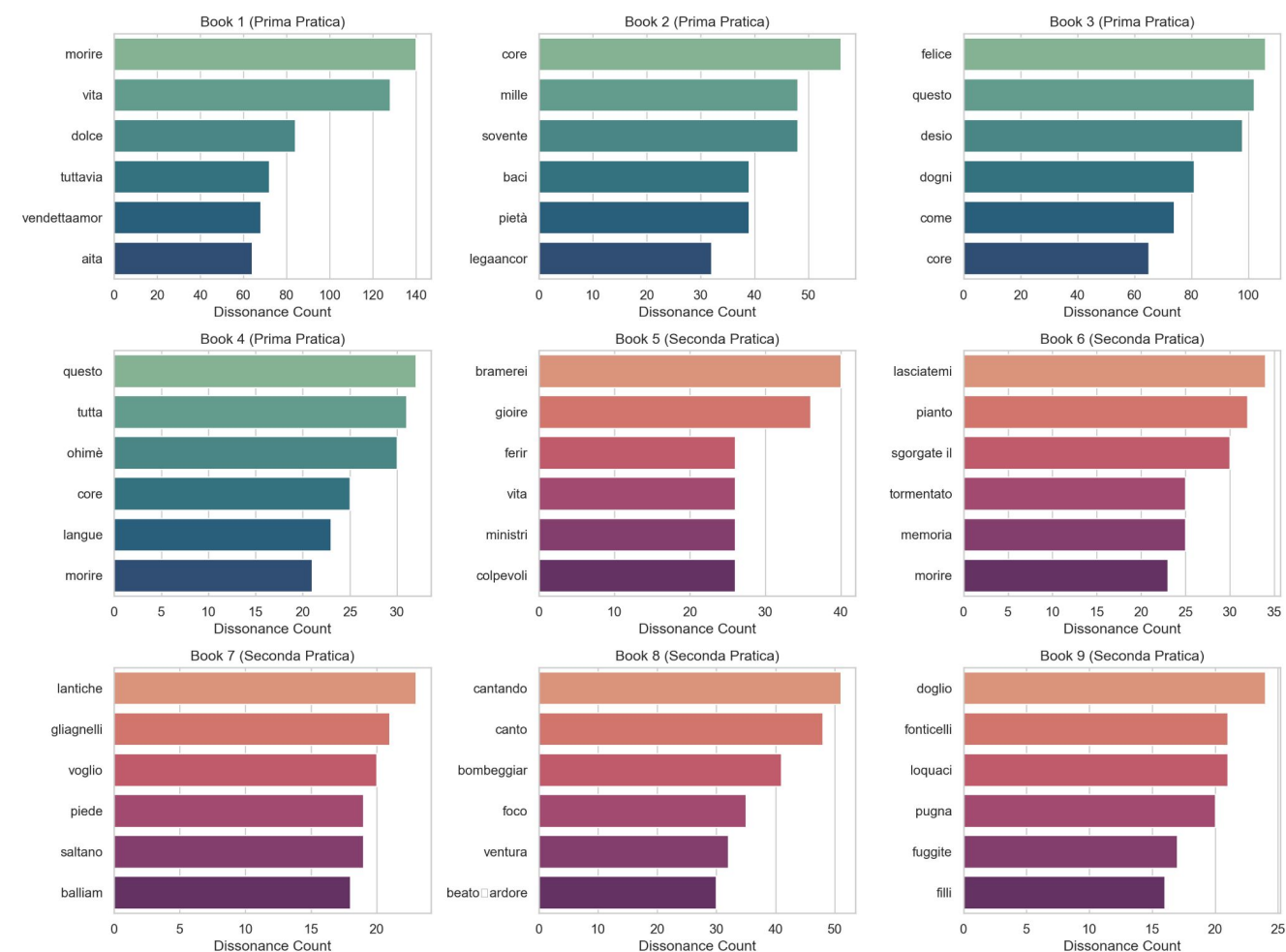
It's also taken into consideration the prepared and the unprepared dissonances.

Dissonant text

Once found the dissonances it can be related directly to the text. Monteverdi began using unprepared and unresolved dissonances specifically on emotionally charged words like *ahi*, *lasso*, *cruda*, and *morire*.



Top 6 Dissonant Words per Book (Full Words Reconstructed)



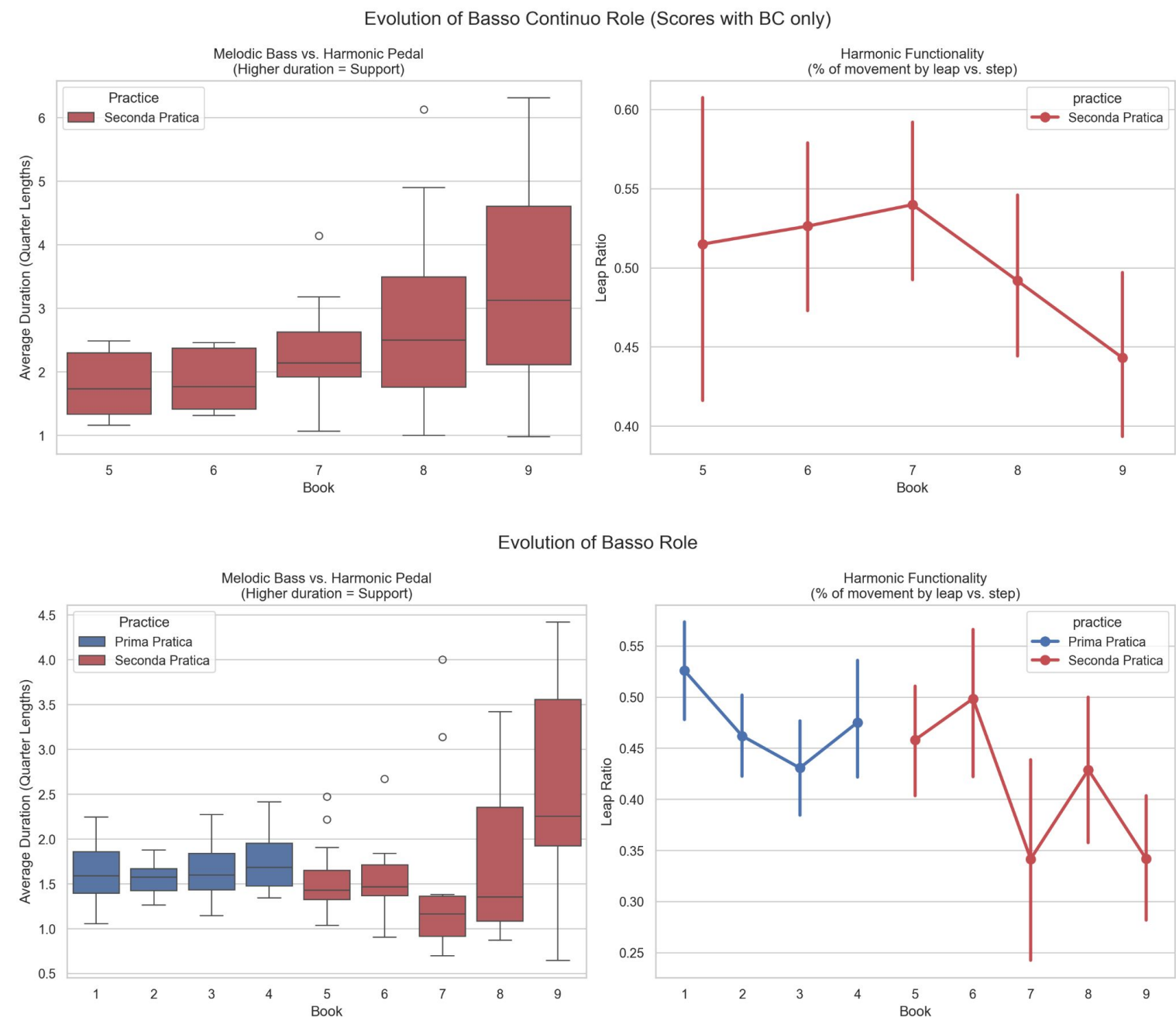
Basso and Basso Continuo

Introduction of the Basso Continuo

The final six madrigals of the fifth book introduce a mandatory, independent instrumental bass line. **Basso Continuo** not only retrieved by the name of the part but also for the sustained bass pedal notes that abandoned polyphony to structurally **support the solo voice above**.

Evolution of Basso

The **bass evolved** from an active, fast-moving melodic voice **into** a **sustained harmonic anchor**, demonstrated by a dramatic increase in note duration and a general decrease in leaping motion in his later books.



Detecting Stile Concitato

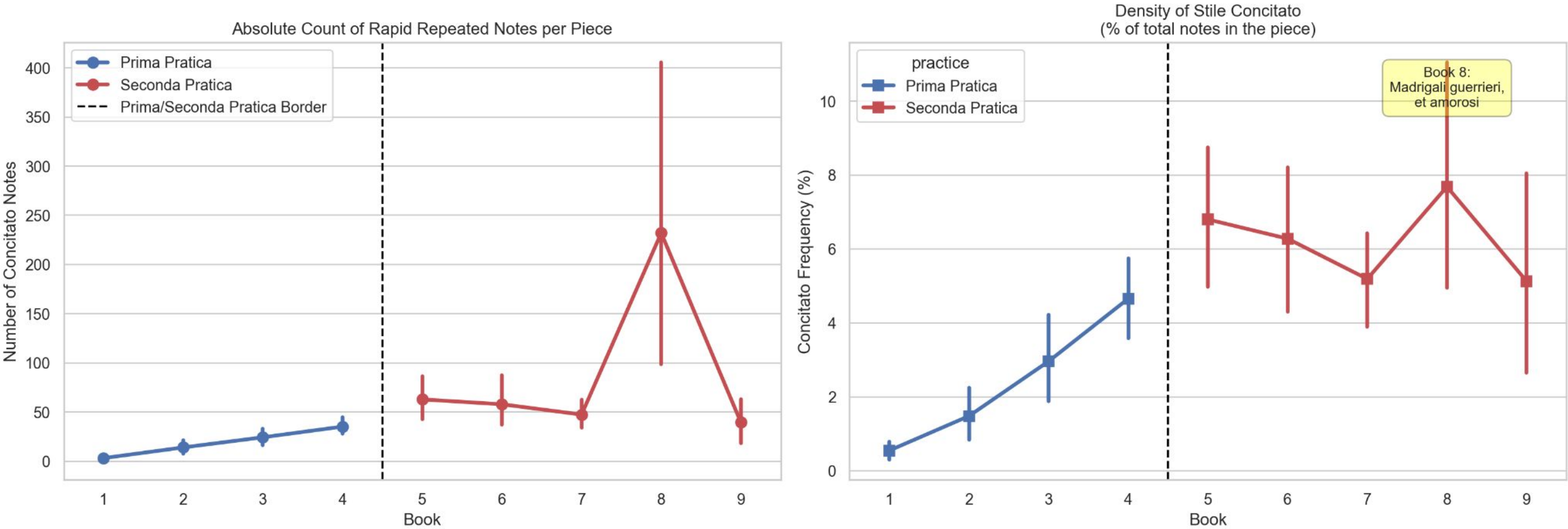
Evolution of "Stile Concitato"

Monteverdi's *stile concitato* (**agitated style**) was invented to express the aggressive passions of war and anger **that earlier music lacked**. He achieved this by dividing a whole note into sixteen rapid, continuously repeated sixteenth notes on a single pitch.

Ogni amante è guerrier, 8th Book

Two staves of musical notation. The top staff is in treble clef and the bottom in bass clef. Both staves show rapid repeated sixteenth notes on a single pitch, characteristic of the stile concitato. The lyrics are: 'ren - ti_e fiu - mi tra piog -' and 'ge e nem - bi varche-ran, varcheran si -'.

Evolution of "Stile Concitato" in Monteverdi's Madrigals



Formalizing **Sprezzatura**

- 📄 **The Core Challenge:** Translating Caccini's "*Tempo dell'animo*" — the soul's expressive freedom — into a computable, reproducible metric.

Syllabic Density Variation

Ratio of rapidly articulated short notes (8th, 16th) against sustained held notes — measuring rhetorical acceleration and expressive elongation.



Syncopation & Beat-Displacement

Frequency of upper voice attacks on **off-beats** while the *basso continuo* anchors strict *Tempo della mano* — quantifying structural rhythmic tension.

Formalizing Sprezzatura



 **Proposal:** Heuristic Sprezzatura Finder.

From the data and the changes analyzed we can find that most of the times Sprezzatura comes from all the calculated features before.



We evaluated each measure on a 0 to 10 scale by summing **three musical heuristics**. The algorithm then filters, ranks, and labels only the measures that score above zero, detailing exactly which features earned their points.

+2

Sustained bass
pedal

+3

Rapid repeated-note
declamation. Stile
Concitato

+5

Lyrics landing on a
dissonant chord

Formalizing Sprezzatura



=====		
PROBABLE MEASURES WITH SPREZZATURA AND ITS FEATURES:		
=====		
Measure	Score	Features
2	10	Bass Pedal/Support Repeated Notes (Declamation) Dissonance on "crurilli"
8	10	Bass Pedal/Support Repeated Notes (Declamation) Dissonance on "col"
9	10	Bass Pedal/Support Repeated Notes (Declamation) Dissonance on "damar, co, col, cora, da, ra, mar"
10	10	Bass Pedal/Support Repeated Notes (Declamation) Dissonance on "co, damar, cora, da, ra, ahi, mar"
13	10	Bass Pedal/Support Repeated Notes (Declamation) Dissonance on "col, las, ra, nome, ahi, che"
14	10	Bass Pedal/Support Repeated Notes (Declamation) Dissonance on "ra, damar"
21	10	Bass Pedal/Support Repeated Notes (Declamation) Dissonance on "segni, se"
33	10	Bass Pedal/Support Repeated Notes (Declamation) Dissonance on "sor, bel, ma, la, più"
34	10	Bass Pedal/Support Repeated Notes (Declamation) Dissonance on "la, sorspido, ma, spido"
37	10	Bass Pedal/Support Repeated Notes (Declamation) Dissonance on "più, de, e, sorda, sor"
38	10	Bass Pedal/Support Repeated Notes (Declamation) Dissonance on "sor, spido, da, e, do, sorda, più"
39	10	Bass Pedal/Support Repeated Notes (Declamation) Dissonance on "fera, ra, e, do, più, sorda, fe"
40	10	Bass Pedal/Support Repeated Notes (Declamation) Dissonance on "più, fugace, ra, fu, fera, sorda"
46	10	Bass Pedal/Support Repeated Notes (Declamation) Dissonance on "cen, mi, fen, i, ta"
51	10	Bass Pedal/Support Repeated Notes (Declamation) Dissonance on "poi, cen, mi, col, dir, rò, mor, ta, morrò, i, che"

Contributions & Significance

Conclusions

Monteverdi's transition from Prima to Seconda Pratica can be **mathematically measured** in terms of:

Harmony: rising unprepared dissonance

Texture: polyphony → homorhythmy

Rhythm: sprezzatura as quantifiable variance

Future Directions

Apply **unsupervised machine learning** (clustering algorithms) to automatically classify a piece as *Prima* or *Seconda Pratica* based purely on its:

- Dissonance profile
- Rhythmic variance signature
- Voice-texture index

Goal: a generalizable classifier applicable beyond Monteverdi.

Thank You for the attention!

Questions?

Project Repository - Scan the QR-Code to visit the Repo!

The GitHub repository contains:

- ChoralWiki MusicXML dataset
- music21 Python analysis scripts
- Statistical output notebooks
- Presentation slides



Course

Analysis of Symbolic Music &
Computational Musicology

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